

Binary Distillation

Pre-Lab Plan

Unit Operations Lab #2

Date: September 17th, 2019

Group 2 of the Tuesday AM section

Evelyn Almonte

Sandra Hernandez

Tianna Mitchell

Anne Serban

Maurice Smith

1. Experimental Objective (5 pts)

The objective of this lab is to examine a binary distillation column to separate ethanol and water and determine certain specifications of the two experiments being done. Experiment A is the Effect of Boil-Up Rate and Overall Column Efficiency. Experiment B is Distillation at Constant Reflux Ratio. For experiment A, the variation with boil-up rate of pressure drop over the distillation column and overall column efficiency is what needs to be determined. For experiment B, the objective will be to carry out a distillation at constant reflux ratios (varying top and bottom compositions with time).

2. Experimental

2.1 Apparatus (5 pts)



Figure 1: Labeled Distillation Apparatus

Table 1: Apparatus Parts and Descriptions

Number	Instrument	Description
1	Distillation Column	Glass distillation column with trays where the VLE is obtained; Separates 10 wt% ethanol-water solution
2	Condenser	Coiled glass tubing where the cooling water flows; Cools the vapor coming out of the distillation column
3	Reboiler	Contains the feed (10 wt% ethanol) and boils feed into the distillation column
4	Heater	Used to heat the reboiler
5	Pump	Pushes feed back into reboiler
6	Funnel to Bottom	Used to charge the reboiler at the start of the experiment
7	Bottom Product Receiver	Collects the bottom product from the reboiler
8	Top Product Receiver	Collects the top product from the decanter
9	Reflux Valve	Controls the reflux within the column
10	Decanter	Decants the product coming out of the condenser into the top product tank
11	Manometer	Used to determine the pressure drop across the column
12	Control Panel	Used to manipulate the automatic valves in the process
13	Thermocouple 1	Monitors the temperature of the uppermost stage in the column
14	Thermocouple 2	Monitors the temperature of the second-uppermost stage in the column
15	Thermocouple 3	Monitors the temperature of the third-uppermost stage in the column
16	Thermocouple 4	Monitors the temperature of the fourth-uppermost stage in the column
17	Thermocouple 5	Monitors the temperature of the fourth-lower stage in the column
18	Thermocouple 6	Monitors the temperature of the third-lower stage in the column
19	Thermocouple 7	Monitors the temperature of the second-lower most stage in the column
20	Thermocouple 8	Monitors the temperature of the bottom stage in the column
21	Valve 6	Located on cooling water pipe to control flow
22	Valve 7	Controls the pressure reading within the column
23	Valve 11	Controls the cooling water entering the condenser
24	Valve 2	Three-way valve that collects sample, closes reboiler, or allows for sample collection
Not Shown in the Diagram		
25	Rotameter	Used to measure the flow of cooling water through condenser
26	Temperature gauge	Used to measure the temperature of the cooling water

2.2 Materials and Supplies (5 pts)

Table 2: Materials and Supplies

Material and Supplies	Purpose
Water	For preparation of 10 L of mixture
Ethanol	For preparation of 10 liters of mixture
Stopwatch	Measure the time required to collect a fixed quantity and calculate flow rate
Graduated Cylinder	To drain condensate
Refractometer	Measure refractive index & composition of samples
Beakers	To collect liquid
Manometer	measure pressure drop
Heat resistant gloves	To protect from high temperatures when handling sample

2.3 Experimental Plan (10 pts)

The lab contains data analysis of experiment A and B. our outline will be divided to two sections. Our goal for experiment A is to vary the boil up ratio to analyze the pressure drop effect. Also we are collecting product samples from the top and bottom of the column to evaluate its efficiency.

Procedure

1. Preparation of solution for refractive index calculations.
 - a. Starting with 9mL of water for 0 wt % solutions, a refractive index was taken.
 - b. Adding 1 mL of ethanol to the water to obtain 10 wt % ethanol solution, another refractive index was taken.
 - c. Increasing amounts of ethanol were added to the solution, to obtain subsequent 20, 30, 40 and 50 wt% refractive indexes
 - d. Procedures a-c were completed again but switching the amount of ethanol and water. This would provide refractive indexes for 100, 90, 80, 70, and 60 wt % refractive indexes.
2. Preparation of ethanol solution for experiment
 - a. Under the assumption that the density of water is 998.2071 g/L at standard laboratory conditions, a 10-L solution of 10% ethanol was prepared.
 - i. Adding 1290g of ethanol to 9L of water, produces a solution that is 10wt% ethanol, that is however 10.3L of solution.
 - ii. After proper mixing of solution, 10L of the solution will be used for the experiment, remaining solution will be discarded using proper MSDS specifications
3. Experimental A: Effect of Boil-Up Rate and Overall Column Efficiency
 - a. Switch reflux ratio timer for operation at total reflux
 - b. Verify that all equipment valves are closed. And double check entire system
 - c. Open valve V10
 - d. Use the funnel (blue hose in the fume hood) to load 10.0 L of the solution through the filler cap at the top of the reboiler. Note: firmly secure cap
 - e. Turn on control panel (power button to the left of panel)
 - f. Set the temperature switch to T9
 - g. Turn on the condense and cooling water by opening valve V5

- i. Leave open until the cooling water flow rate F11 to the condenser is approximately 3.0 L/min
 - h. From the control panel, turn on the Reboiler Power Control to 1.5 kW
 - i. Switch on the heating element power
 - ii. Record the temperatures per minute as steady state is reached
 - i. Open Valve V6, then V7
 - j. Close V6 and V7 which shouldn't have an initial pressure difference.
 - k. Note the appearance of the first and fifth tray
 - i. T1 through T8 should change as the process begins
 - ii. Vapor will form droplets once condensed
 - iii. The system will run under total reflux
 - l. After steady bubbling is established, turn the power controller clockwise until it reads 0.50 kW
 - m. Once all the temperatures from T1-T8 are constant, the system has reached equilibrium. Record the temperatures
 - n. Measure the boil up rate at 0.50 kW by recording time-volume collection
 - i. Use V3 and drain the flexible pipe to reach steady flow out.
 - ii. Use a stopwatch to record set volume
 - iii. Secure liquid in flexible tube to prevent vapor from escaping
 - iv. Save the solution for measurement then discard after
 - o. Take readings pressure drops at top and bottom (V6 then V7 rare opened respectively).
 - i. Record refractive index
 - ii. Repeat procedure for three samples of top and bottom product and record temperature T1-T8
 - p. Step up boil-up rate in 250 Watts increments to a maximum of 1.25 kW and take similar readings.
4. Experiment B: Distillation at Constant Reflux Ratio
 - a. While solution from Experiment A is still inside of the column, set the heat controller to 75 kW, then leave until equilibrium conditions are achieved.
 - b. Set the reflux ratio of the solution, done by changing the time back to column and time to top product receiver.
 - c. Switch the reflex valve on.
 - d. Open valve V3 and collect the condensate into a measuring cylinder.
 - i. Set aside first sample collected to be discarded, roughly 5-10 mL
Removal of too much solution disturbs the overall mass balance.
 - ii. Partially open valve V3 and drain from the reflex until steady state is obtained, careful to leave a small amount of liquid in the line at all-times.
 - iii. Collect subsequent sample into measuring sample, this time recording how long it takes to collect a set quantity.
Repeat collection of condensates 3 times to determine average value of boil-up rate.
 - e. Take refractive index for taken overhead sample.
 - f. While completing step f, take a sample of the bottom through valve V2.
 - g. Record refractive index of solution obtained through bottom.
 - h. Repeat procedure every 10 minutes until 5 samples of both overhead and bottom are obtained.

Record temperatures T8 and T1 to calculate the average column temperature

- i. Complete entire procedure again for at least one different reflux ratio.



Project Safety Evaluation (10 pts)

Project Title: Distillation

Date: Tuesday, September 17th, 2019

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	If yes, identify conditions: Do not exceed required voltage, and be cautious with liquids around the electrical components
Potential mechanical hazards?	Yes	If yes, identify conditions: Distillation column, pump, potential falling hazards from heights greater than 6 feet.
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	Yes	If yes, identify conditions: The sample being taken from the bottom will be hot, temperature resistant gloves are to be worn when collecting solution.
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: Flammable liquid, fatal if swallowed. If ingested or high concentration inhaled, may cause drowsiness, unconsciousness and central nervous system depression.
Gloves required?	Yes	If yes, specify type: Use Latex gloves while preparing and pouring the ethanol solution and heat resistant gloves when collecting samples
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Unit Operations Lab MSDS, <i>Ethyl Alcohol</i>
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	130C	
Pressure (psia)	60psig	
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
Safety attire	yes	Wear gloves, goggles, and proper clothing. Also wear heat resistant gloves when handling the samples

I have read relevant background material for the Unit Operations Laboratory entitled: "Distillation" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

1. M Heel

signature of team member

09/17/19

date

Evelyn Amante

signature of team member

9/17/19

date

[Signature]

signature of team member

9-17-19

date

[Signature]

signature of team member

9/17/19

date

[Signature]

signature of team member

09.17.19

date